

Classes 3-5

Why doesn't the size of the House change anymore? [See how House would have to scale with the population over time at this interactive Census applet](#)¹

Apportionment is rife with paradoxes. Let's dig in.

That's not fair!

Example 1 In our CJ Coffee Distribution Scheme, there were some possibilities we didn't consider.

Table 2 Hamilton Apportionment

Prof.	A	B	Ce	Co	D	G	Ka	Ku	L	T	Total
Pop	35	60	71	22	38	55	73	45	28	58	485
Q	3.25	5.57	6.59	2.04	3.53	5.10	6.77	4.18	2.60	5.38	~485
Boxes	5	5	5	5	5	5	5	5	5	0	45
Boxes	4	4	4	4	4	9	4	4	4	4	45

□

The Apportionment Problem, but Mathier

Formal Mathematical Variables

The problem can be defined using the following parameters:

The (Exact) Quota

A central concept in apportionment is the **exact quota**, denoted q_s , which represents the ideal fractional share of units an entity would receive if units were divisible. It is defined by

$$q_s = \frac{p_s}{P} M.$$

In an ideal mathematical world, we would have $m_s = q_s$. However, because m_s must be an integer and q_s is typically a rational number that is not an integer, an apportionment method is required to map the real vector of quotas to an integer vector of allocations.

Definition 3 Apportionment Method: a function! Fix a positive integer M and an integer $n \geq 2$. Let $\mathcal{Q} = \{\mathbf{q} \in \mathbb{R}_{\geq 0}^n : \sum_{s=1}^n q_s = M\}$ denote the space of exact quota vectors, and let $\mathcal{M} = \{\mathbf{m} \in \mathbb{Z}_{\geq 0}^n : \sum_{s=1}^n m_s = M\}$ denote the set of feasible integer allocations.

An **apportionment method** is a function

$$F : \mathcal{Q} \longrightarrow \mathcal{M}$$

that assigns to each exact quota vector $\mathbf{q}$ a unique integer allocation $\mathbf{m} = F(\mathbf{q})$.

The method F induces a partition of $\mathcal{Q}$ into regions

$$P(\mathbf{m}) = \{\mathbf{q} \in \mathcal{Q} : F(\mathbf{q}) = \mathbf{m}\},$$

called the **apportionment polyhedra**. Each polyhedron consists of all exact quota vectors that are mapped to the same integer allocation. ◇

¹public.tableau.com/views/Apportionment_16116063135030/Apportionment?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

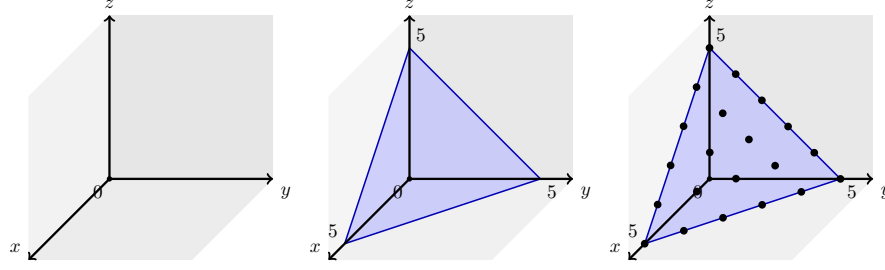


Figure 4 The positive orthant, $\mathcal{Q}$, and $\mathcal{M}$ with three states and five seats.

Apportionment Polyhedra

An **apportionment polyhedron** is a geometric region within the space of all possible exact quota vectors $\mathbf{q}$ that a particular apportionment method maps to a single, fixed integer seat allocation $\mathbf{m}$. In mathematical terms, it consists of all fractional quota vectors that result in the same whole-number distribution of seats.

Geometric Representation

The concept is best understood by visualizing the space of exact quotas.

- *The set $\mathcal{Q}$.* This set contains all possible fractional distributions of M items among n entities. For example, if there are three parties and $M = 5$ seats, the set $\mathcal{Q}$ forms an equilateral triangle in three-dimensional space with vertices at $(5, 0, 0)$, $(0, 5, 0)$, and $(0, 0, 5)$.
- *Integer points.* Within this geometric space, the possible integer seat allocations are represented by discrete lattice points corresponding to integer vectors $\mathbf{m}$.
- *Defining the polyhedra.* A valid apportionment method partitions the larger region $\mathcal{Q}$ into smaller regions, or polyhedra, each associated with a single integer allocation. Any exact quota vector lying inside a given polyhedron is rounded or mapped to the integer allocation corresponding to that region.

The shape and size of apportionment polyhedra depend on the error function or rounding rule associated with the chosen apportionment method.

- *Mapping.* A desirable property of an apportionment method is that any exact quota vector that is already an integer vector is mapped to itself.
- *Borders.* The boundaries between apportionment polyhedra correspond to tie situations, where an exact quota vector may be equally compatible with two or more integer seat allocations. In such cases, the method must include a rule for resolving ties along these boundaries.

To find apportionment polyhedra given an apportionment function, it is useful to introduce **barycentric coordinates**. Given a value M , a barycentric n -tuple $(q_1, q_2, \dots, q_m)$ satisfies the property that $q_i \geq 0$ for each coordinate and $\sum_{i=1}^n q_i = M$. We can imagine each tuple with an M in one spot and zeros elsewhere to be sitting at the corner of an n -dimensional simplex, which results from treating the tuple like the end of a vector with tail at the original in $\mathbb{R}_{\geq 0}^n$. Given a coordinate q_i , the face opposite it is the set of all barycentric coordinates with a zero in position i .

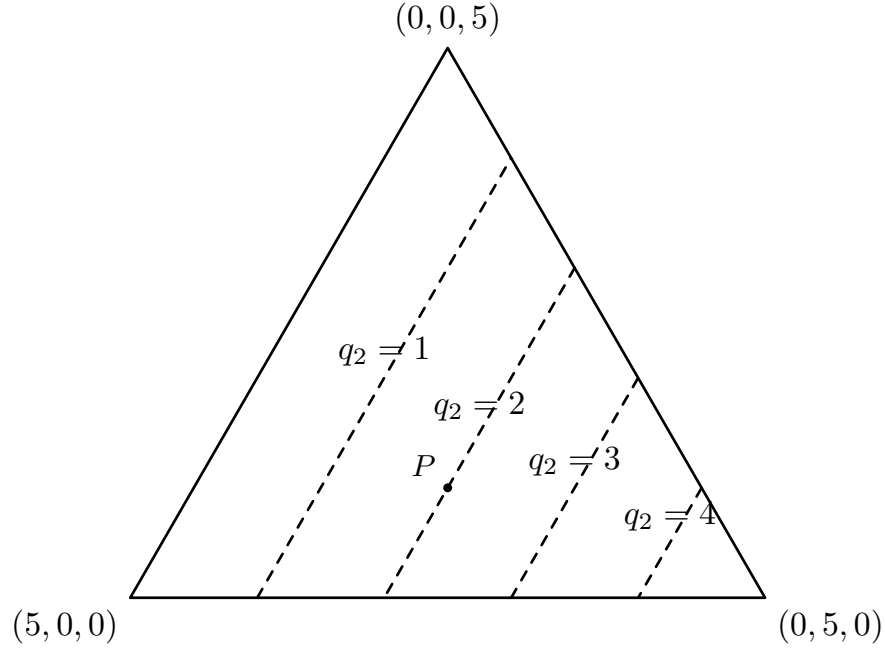


Figure 5 Moving around the triangle where the barycentric coordinates sum to five. The barycentric tuple $P = (2, 2, 1)$ lies on the line $q_2 = 2$ (and the line $q_1 = 2$, and the line $q_3 = 1$).

Mathematical Constraints and Properties

For an apportionment method to be considered fair or valid, it is often expected to satisfy certain mathematical axioms:

- *Symmetry.* The outcome should not depend on the order in which the entities are listed.
- *Homogeneity.* If all input values q_j are multiplied by the same positive constant, the resulting apportionment should remain unchanged.
- *The Quota Rule.* A method satisfies this rule if each allocation m_j is either $\lfloor q_j \rfloor$ or $\lceil q_j \rceil$.
- *House Monotonicity.* If the total number of units increases from M to $M + 1$, no entity's allocation m_j should decrease. This property rules out paradoxes such as the Alabama Paradox.

The Balinski-Young Theorem essentially says that there is no apportionment rule that satisfies all of these criteria all the time!

Hamilton's Method as an Apportionment Function

Fix M seats and n entities. Given an exact quota vector $\mathbf{q} \in \mathcal{Q}$, Hamilton's method proceeds as follows:

1. Assign each entity its lower quota $m_j = \lfloor q_j \rfloor$.
2. Distribute the remaining seats, one at a time, to the entities with the largest fractional remainders $q_j - \lfloor q_j \rfloor$.

This defines an apportionment method $F_{\text{Ham}} : \mathcal{Q} \rightarrow \mathcal{M}$, since every quota vector is mapped to a unique integer allocation.

Geometrically, the apportionment polyhedra for Hamilton's method are regions determined by inequalities comparing fractional parts of the quotas. The boundaries between polyhedra occur where two or more remainders are equal, leading to ties in the final seat assignment.

Because these boundaries depend on global comparisons of remainders, the resulting polyhedra need not arise from minimizing a single global error function. This explains why Hamilton's method can satisfy the Quota Rule while still exhibiting paradoxes such as the Alabama Paradox.

Example 6 The Hamilton Polyhedron for an example.. Suppose we have $M = 15$ boxes of coffee to distribute among three people (Ahmed, Gibbons, and Tripp) in proportion to their "need scores" $\mathbf{a} = (35, 55, 58)$.

The total is $A = 35 + 55 + 58 = 148$, so the exact quotas are $q_j = \frac{a_j}{A}M$.

Table 7 Exact quotas and remainders for 15 boxes

Person	a_j	$q_j = \frac{a_j}{148} \cdot 15$	$\lfloor q_j \rfloor$	Remainder
Ahmed	35	$\frac{525}{148} \approx 3.547$	3	≈ 0.547
Gibbons	55	$\frac{825}{148} \approx 5.574$	5	≈ 0.574
Tripp	58	$\frac{870}{148} \approx 5.878$	5	≈ 0.878

Using Hamilton's (largest remainder) method, we first assign lower quotas: $(3, 5, 5)$, which totals 13 boxes, leaving 2 boxes to distribute. The two largest remainders are Tripp's and Gibbons's, so each receives one additional box.

The final allocation is

$$(m_{\text{Ahmed}}, m_{\text{Gibbons}}, m_{\text{Tripp}}) = (3, 6, 6).$$

Consider Hamilton's method on the quota simplex $\mathcal{Q} = \{(q_A, q_G, q_T) \in \mathbb{R}_{\geq 0}^3 : q_A + q_G + q_T = 15\}$. We describe the region of quota vectors that Hamilton maps to the allocation $(3, 6, 6)$.

For Hamilton's method to output $(3, 6, 6)$, the lower-quota step must first produce

- Two seats left: $(2, 5, 6)$, $(2, 6, 5)$, or $(3, 5, 5)$;
- One seat left: $(2, 6, 6)$, $(3, 5, 6)$, or $(3, 6, 5)$;
- No seats left: $(3, 6, 6)$.

In our example, we have the scenario that the lower quotas produce $(3, 5, 5)$, and the remainders satisfy $r_G > r_A$, $r_T > r_A$:

$$3 \leq q_A < 4, \quad 5 \leq q_G < 6, \quad 5 \leq q_T < 6, \quad q_A + q_G + q_T = 15.$$

Write the remainders as $r_A = q_A - 3$, $r_G = q_G - 5$, and $r_T = q_T - 5$. To award the two remaining seats to Gibbons and Tripp (and not to Ahmed), we need the two inequalities

$$r_G > r_A, \quad r_T > r_A.$$

Equivalently, these are linear conditions on the quotas:

$$q_G - q_A > 2, \quad q_T - q_A > 2.$$

Therefore, part of the (open) Hamilton region mapping to $(3, 6, 6)$ is

$$P(3, 6, 6) \supseteq \left\{ (q_A, q_G, q_T) \in \mathcal{Q} : \begin{array}{l} 3 \leq q_A < 4, \ 5 \leq q_G < 6, \ 5 \leq q_T < 6, \\ q_G - q_A > 2, \ q_T - q_A > 2 \end{array} \right\}.$$

The boundary of this region consists of tie situations, such as $r_G = r_A$ or $r_T = r_A$, together with the grid boundaries where a quota crosses an integer and a lower quota changes (for example, $q_G = 6$). A complete closed polyhedron requires a specified tie-breaking rule on these boundary sets. $\square$